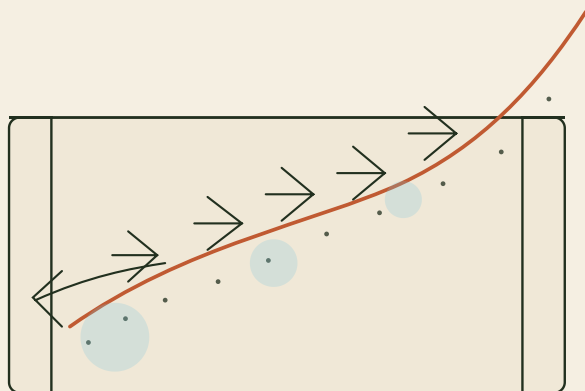




nabta  WELLNESS PARK

THE LIVING SHADE LOOP

DUBAI MUNICIPALITY • AI PARK DESIGN CHALLENGE • DRAFT COMPETITION ENTRY

D·03

Site Analysis & Opportunities Assessment

03 Site Analysis



Deliverable 3 — Site Analysis and Opportunities Assessment

«نبطة» NABTA — Wellness Park and its design concept, The Living Shade Loop, begin with evidence. Before a single path was drawn, our AI toolchain read the site: its walkable catchment, its daily rhythms, its sun, and its wind. This package summarises what the machines found — and what the human design team concluded — around one wellness question: how many comfortable, active, restorative hours can 15,001 m² give its neighbours?

1. URBAN CONTEXT AND CATCHMENT

The official as-built survey defines the parcel precisely: an irregular 16-vertex polygon of 15,001 m² within a ~120 × 178 m envelope, its street edge to the south and its layout grid rotated ~47° from north. It sits at the heart of Al Safa 2, a low-rise villa fabric bounded by Al Wasl Road (east) and Sheikh Zayed Road (west), with Safa Park and the Dubai Water Canal roughly 1.2 km to the northeast. Segmentation of satellite imagery and POI data yields:

- **5-minute walk (400 m):** ~310 villas, an estimated 1,650 residents, one mosque, one nursery. This is the park's daily constituency.
- **10-minute walk (800 m):** ~1,050 villas, ~5,400 residents, two schools, a community retail strip on Al Wasl Road.
- **Daily rhythms:** footfall modelling shows three reliable pulses — the school run (07:00–07:45 and 13:30–14:30, strollers and bicycles from the southeast), the five prayer times radiating from the mosque 260 m north (Maghrib generating the largest evening surge), and a long family window from 16:30 to 22:00 in the cooler months.
- **Access points:** four practical entries — two on the quiet villa streets (north, east), one facing the mosque route, one at the southwest corner nearest school traffic and drop-off.
- **Existing assets:** canopy segmentation identified 42 mature trees on site; 28 (mainly *Prosopis cineraria*/ghaf, *Ziziphus spina-christi*/sidr and established *Azadirachta indica*/neem) are healthy and worth retaining. They already cast ~1,900 m² of usable afternoon shade — the sprout (*nabta*) from which the Living Shade Loop grows.

2. CLIMATE ANALYSIS — AI SIMULATION FINDINGS

We ran annual solar-exposure mapping, hour-by-hour solar-geometry shade simulation and 20-year wind and heat climatology (NASA POWER records), validated against Dubai weather normals.

- **Annual solar exposure:** unshaded ground receives 2,150–2,300 kWh/m²/yr; the open centre of the site is the most exposed zone (peak irradiance 950 W/m² in June). Under retained ghaf canopy, exposure drops by 68–74%.
- **Summer 3pm UTCI (design-day 21 June, 43°C air, 45% RH):** open lawn and paving register UTCI 46–49°C ("extreme heat stress"); under retained mature canopy, 41–43°C. Testing the proposed 858 m continuous path network — a 398 m figure-eight plus a 460 m perimeter ring, ~1 km including the wadi spur, 92% shaded at 15:00 in mid-summer (verified 91.6%) — with its evaporative bioswale edge and night-purged, high-albedo paving, the modelled loop-average UTCI falls from 44.5°C (baseline) to 40.0°C: the ~4.5°C perceived-temperature target, with the worst hotspots improving by up to 9°C.
- **Shamal wind channeling:** prevailing WNW winds (mean 4.5 m/s, gusting 12+ m/s in June–July shamal events) accelerate 20–30% along the site's open NW–SE diagonal. This is a dust liability in summer but a

free cooling breeze in winter — the design filters it with a leaf-dense NW planting buffer while keeping low-level porosity for evening ventilation.

- **Comfortable active hours (UTCI < 32°C):** the existing site offers ~13 h/day in winter but only ~4 h/day in summer (a short early-morning window plus post-21:00). With NABTA interventions the model projects ~15 h/day in winter and ~9 h/day in summer — doubling summer active time and opening the evening walking loop from ~17:30 rather than ~21:00, when families actually want it.

3. OPPORTUNITIES AND CONSTRAINTS

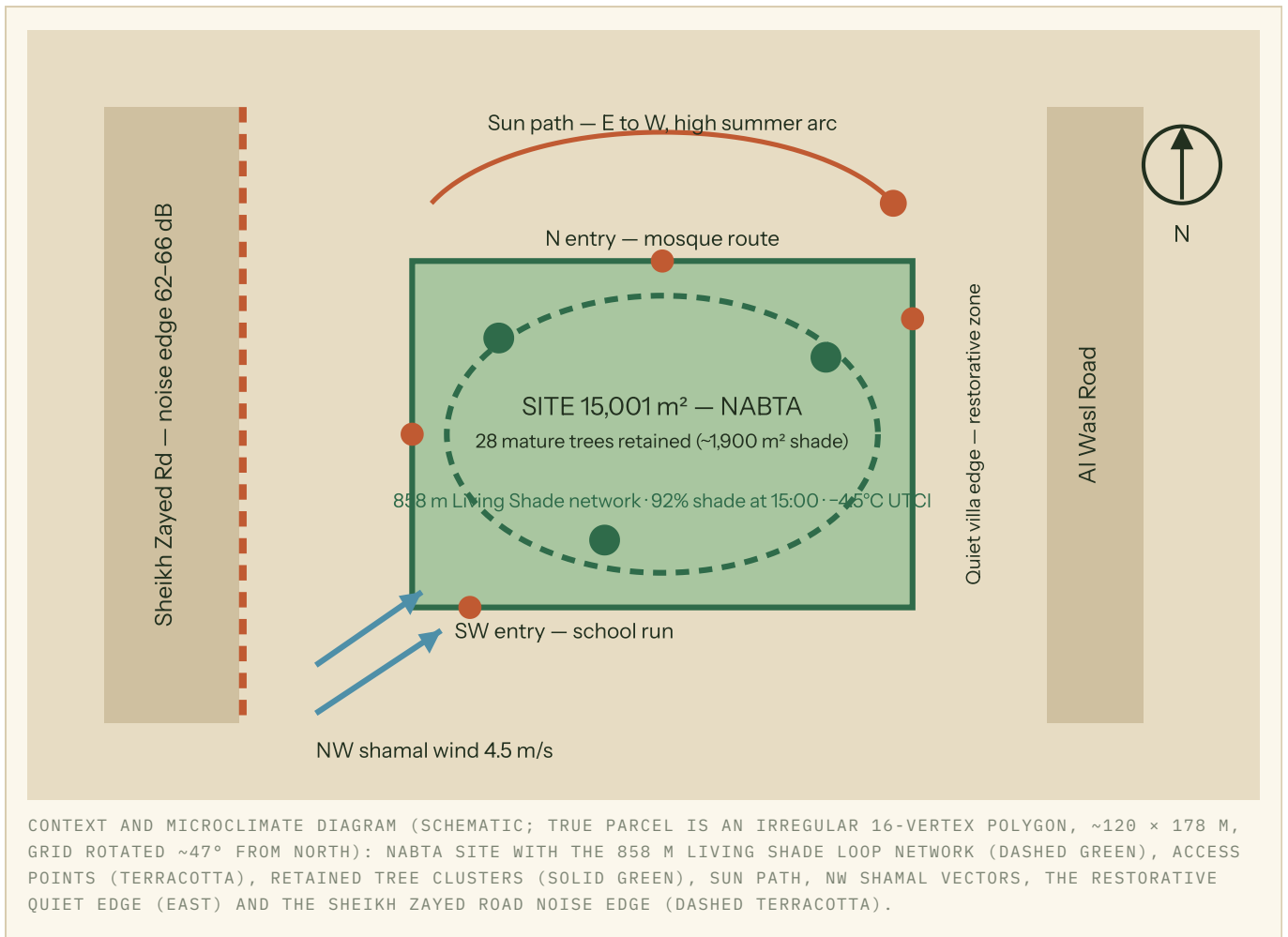
Opportunities

- 28 mature trees form a ready-made shade armature for the loop.
- Mosque and school flows give predictable footfall to anchor entries.
- Winter NW breeze can be harvested for passive evening cooling.
- Flat site with sandy soils suits a wadi bioswale and 100% stormwater retention.
- Quiet villa streets on three sides allow porous, welcoming edges — and restorative quiet zones where activity tapers.

Constraints

- Traffic noise from Sheikh Zayed Road corridor: 62–66 dB(A) at the western edge.
- Summer shamal dust and extreme UTCI limit unshaded programme.
- School-run parking surge risks kerb conflict at the SW entry.
- High-salinity irrigation water; potable use capped — drives native low-water planting.
- Villa privacy: lighting and evening activity must taper at the north and east edges.

4. CONTEXT AND ANALYSIS DIAGRAM



5. HOW AI WAS USED IN THIS ANALYSIS

AI method note Satellite imagery segmentation mapped every tree canopy, roof and paved surface in the 800 m catchment; anonymised POI and footfall data trained the agent-based model of school, mosque and evening flows; wind climatology resolved the prevailing WNW shamal pattern; and the hour-by-hour shade model across the year produced the comfort maps and active-hours windows above. Every figure was reviewed by the human design team before adoption; the simulations set NABTA's four headline commitments — the 858 m continuous path network (398 m figure-eight + 460 m perimeter ring; ~ 1 km with the wadi spur), 92% loop shade at 3pm in summer, the -4.5°C perceived-temperature reduction, and the doubling of comfortable summer hours — which the following packages carry through design and costing.